

MYSQL Assignment

1. Car policy details (policytypeid, policytypename, description)

```
SELECT ps.policy_type_id,  
rpt.policy_type_name,  
ps.description  
FROM policy_sub_types ps  
JOIN ref_policy_types rpt  
ON ps.policy_type_code = rpt.policy_type_code  
WHERE rpt.policy_type_name = 'car';
```

2. Policy type code & number of policies

```
SELECT policy_type_code,  
COUNT(*) AS NO_OF_POLICIES  
FROM policy_sub_types  
GROUP BY policy_type_code;
```

3. Users residing in Chennai

```
SELECT u.user_id, u.firstname, u.lastname, u.email, u.mobilen  
FROM user_details u  
JOIN address_details a  
ON u.address_id = a.address_id  
WHERE a.city = 'chennai';
```

4. Users who have taken car policies

```
SELECT u.user_id,  
CONCAT(u.firstname, ' ', u.lastname) AS USER_NAME,  
u.email, u.mobilen  
FROM user_details u  
JOIN user_policies up ON u.user_id = up.user_id  
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id  
JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code  
WHERE rpt.policy_type_name = 'car';
```

5. Users who took car but not home policies

```
SELECT DISTINCT u.user_id, u.firstname, u.lastname  
FROM user_details u  
JOIN user_policies up ON u.user_id = up.user_id  
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id  
WHERE ps.policy_type_code = '58934'  
AND u.user_id NOT IN (
```

```
SELECT u2.user_id
FROM user_details u2
JOIN user_policies up2 ON u2.user_id = up2.user_id
JOIN policy_sub_types ps2 ON up2.policy_type_id = ps2.policy_type_id
WHERE ps2.policy_type_code = '58539'
);
```

6. Policy type with maximum number of policies

```
SELECT rpt.policy_type_code, rpt.policy_type_name
FROM ref_policy_types rpt
JOIN policy_sub_types ps ON rpt.policy_type_code = ps.policy_type_code
GROUP BY rpt.policy_type_code, rpt.policy_type_name
ORDER BY COUNT(*) DESC
LIMIT 1;
```

7. Users whose city ends with 'bad'

```
SELECT u.user_id, u.firstname, u.lastname, a.city, a.state
FROM user_details u
JOIN address_details a
ON u.address_id = a.address_id
WHERE a.city LIKE '%bad';
```

8. Users registered before May 2012

```
SELECT u.user_id, u.firstname, u.lastname,
up.policy_no, up.date_registered
FROM user_details u
JOIN user_policies up
ON u.user_id = up.user_id
WHERE up.date_registered < '2012-05-01';
```

9. Users with more than one policy

```
SELECT user_id
FROM user_policies
GROUP BY user_id
HAVING COUNT(policy_no) > 1;
```

10. Policies maturing in August 2013

```
SELECT rpt.policy_type_code, rpt.policy_type_name,
ps.policy_type_id, up.user_id, up.policy_no
FROM user_policies up
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id
JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code
WHERE DATE_ADD(up.date_registered, INTERVAL ps.maturityperiod YEAR)
BETWEEN '2013-08-01' AND '2013-08-31';
```

11. Maturity amount > double total paid

```
SELECT ps.policy_type_code, rpt.policy_type_name, ps.policy_type_id
FROM policy_sub_types ps
JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code
JOIN policy_payments pp ON ps.policy_type_id = pp.policy_no
GROUP BY ps.policy_type_id, ps.maturityamount
HAVING ps.maturityamount > 2 * SUM(pp.amount);
```

12. Total amount paid by each user

```
SELECT user_id,
SUM(amount) AS total_amount
FROM policy_payments
GROUP BY user_id;
```

13. Total paid for each policy

```
SELECT user_id, policy_no,
SUM(amount) AS total_amount
FROM policy_payments
GROUP BY user_id, policy_no;
```

14. Balance amount for each policy

```
SELECT up.user_id, up.policy_no,
ps.amount * ps.yearsofpayments - SUM(pp.amount) AS balance_amount
FROM user_policies up
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id
JOIN policy_payments pp ON up.policy_no = pp.policy_no
GROUP BY up.user_id, up.policy_no;
```

15. Balance payment years

```
SELECT up.user_id, up.policy_no,
ps.yearsofpayments - COUNT(pp.receipno) AS BALANCE_YEARS
FROM user_policies up
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id
LEFT JOIN policy_payments pp ON up.policy_no = pp.policy_no
GROUP BY up.user_id, up.policy_no, ps.yearsofpayments;
```

16. Users who took car, home & life policies

```
SELECT u.user_id, u.firstname, u.lastname
FROM user_details u
JOIN user_policies up ON u.user_id = up.user_id
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id
GROUP BY u.user_id
HAVING COUNT(DISTINCT ps.policy_type_code) = 3;
```

17. Total amount paid per policy department

```
SELECT ps.policy_type_code,  
SUM(pp.amount) AS total_amount  
FROM policy_sub_types ps  
JOIN policy_payments pp ON ps.policy_type_id = pp.policy_no  
GROUP BY ps.policy_type_code;
```

18. Users with multiple policy types under same policy code

```
SELECT up.user_id,  
CONCAT(u.firstname,' ',u.lastname) AS user_name,  
ps.policy_type_code,  
ps.policy_type_id  
FROM user_policies up  
JOIN user_details u ON up.user_id = u.user_id  
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id  
GROUP BY up.user_id, ps.policy_type_code  
HAVING COUNT(DISTINCT ps.policy_type_id) > 1;
```

19. Policy department with minimum policies

```
SELECT rpt.policy_type_code, rpt.policy_type_name  
FROM ref_policy_types rpt  
JOIN policy_sub_types ps ON rpt.policy_type_code = ps.policy_type_code  
GROUP BY rpt.policy_type_code  
ORDER BY COUNT(*) ASC  
LIMIT 1;
```

20. Users who completed all payments

```
SELECT u.user_id,  
CONCAT(u.firstname,' ',u.lastname) AS user_name,  
a.city AS address,  
u.mobileno,  
ps.policy_type_code,  
ps.policy_type_id,  
rpt.policy_type_name  
FROM user_details u  
JOIN address_details a ON u.address_id = a.address_id  
JOIN user_policies up ON u.user_id = up.user_id  
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id  
JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code  
JOIN policy_payments pp ON up.policy_no = pp.policy_no  
GROUP BY up.policy_no  
HAVING COUNT(pp.receipno) = ps.yearsofpayements;
```

21. Latest 2 registered users

```
SELECT u.user_id,  
CONCAT(u.firstname,' ',u.lastname) AS user_name,  
a.city AS address,
```

```
u.mobileno,  
ps.policy_type_code,  
ps.policy_type_id,  
rpt.policy_type_name,  
up.date_registered  
FROM user_details u  
JOIN address_details a ON u.address_id = a.address_id  
JOIN user_policies up ON u.user_id = up.user_id  
JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id  
JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code  
ORDER BY up.date_registered DESC  
LIMIT 2;
```
